Daily Logic Drill #1 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

Answers and Methods.

Section A Rapid Recognition

1. Identify the hypothesis and the conclusion of: "If n is divisible by 10, then n is divisible by 5."

Answer: Hypothesis: n is divisible by 10. Conclusion: n is divisible by 5.

Method: The clause following "if" is the hypothesis; the clause following "then" is the conclusion.

> **Investigate further:** Every multiple of 10 is a multiple of 5 — is the converse ("multiple of 5 $\Rightarrow$ multiple of 10") also true? Test $n = 15$.

2. State the converse of: "If $x = 3$, then $x^2 = 9$."

Answer: "If $x^2 = 9$, then $x = 3$."

Method: Swap the hypothesis and conclusion of the original conditional: the hypothesis $x = 3$ becomes the converse's conclusion, and the conclusion $x^2 = 9$ becomes the converse's hypothesis.

> **Investigate further:** This converse is false — $x = -3$ also satisfies $x^2 = 9$. Section C's counterexample-hunting formalises exactly this kind of check.

3. True or false: "If a shape is a square, then it is a rectangle."

Answer: True.

Method: A square has four right angles, which is precisely the defining property of a rectangle; a square is simply a rectangle with the extra constraint of equal side lengths.

> **Investigate further:** Is the converse (A4) also true?

4. True or false: "If a shape is a rectangle, then it is a square."

Answer: False.

Method: A 2×3 rectangle has four right angles but unequal side lengths, so it is not a square. A single such example is enough to disprove a universal "if...then" claim.

> **Investigate further:** A3 was true but its converse (A4) is false — this asymmetry is the default, not the exception.

5. State the converse of: "If n is even, then n^2 is even."

Answer: "If n^2 is even, then n is even."

Method: Swap hypothesis and conclusion: the new hypothesis is "n^2 is even", the new conclusion is "n is even".

> **Investigate further:** Unlike A2's converse, this one is also true (see B2 and B5 for the direct proofs of both directions).

6. True or false: "$n^2 > 0$ for every integer n."

Answer: False.

Method: The integer $n = 0$ gives $n^2 = 0$, which is not greater than 0. One counterexample disproves a "for every" claim.

> **Investigate further:** Restrict to nonzero integers: is $n^2 > 0$ then true for every $n \neq 0$?

7. In "If $x > 3$ then $x^2 > 9$", which part is the hypothesis and which is the conclusion?

Answer: Hypothesis: $x > 3$. Conclusion: $x^2 > 9$.

Method: The clause following "if" is the hypothesis; the clause following "then" is the conclusion.

> **Investigate further:** Is the converse "$x^2 > 9 \Rightarrow x > 3$" true? Test $x = -4$.

8. True or false: "If $a = b$, then $a^2 = b^2$."

Answer: True.

Method: If $a = b$, substituting a for b in a^2 gives $a^2 = a \cdot a = b \cdot b = b^2$ directly.

> **Investigate further:** Is the converse "$a^2 = b^2 \Rightarrow a = b$" true? Test $a = 2, b = -2$.

9. State the converse of: "If n is a multiple of 6, then n is a multiple of 2."

Answer: "If n is a multiple of 2, then n is a multiple of 6."

Method: Swap hypothesis and conclusion.

> **Investigate further:** This converse is false ($n = 4$ is a multiple of 2 but not 6) — Day 4's toolkit formalises this counterexample search.

10. True or false: "The converse of a true statement is always true."

Answer: False.

Method: A2 and A9 each give an explicit true statement whose converse is false. One such example disproves the universal claim "always."

> **Investigate further:** You've now seen both patterns: true statements with false converses (A2, A9) and, per A5, true statements with true converses. Section B starts building a feel for telling them apart.

Section B Manipulation Drills

1. Write the converse of: "If n is divisible by 4 and by 3, then n is divisible by 12." State whether the original statement and its converse are both true.

Answer: Converse: "If n is divisible by 12, then n is divisible by 4 and by 3." Both the original and the converse are true.

Method: Original: if $4 \mid n$ and $3 \mid n$, then since $\gcd(4, 3) = 1$, $\operatorname{lcm}(4, 3) = 12$ divides n. Converse: if $12 \mid n$, then since $4 \mid 12$ and $3 \mid 12$, both $4 \mid n$ and $3 \mid n$ follow directly.

> **Investigate further:** When a statement and its converse are both true, they can be joined with "if and only if" — exactly Day 2's new tool.

2. Prove directly: "If n is an even integer, then n^2 is even."

Answer: Proved: writing $n = 2k$ gives $n^2 = 4k^2 = 2(2k^2)$, an even number.

Method: Let n be even, so $n = 2k$ for some integer k. Then $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$, which is twice an integer, hence even.

> **Investigate further:** Compare with B5: does the converse "n^2 even $\Rightarrow n$ even" also hold?

3. State the converse of: "If $x > 0$ and $y > 0$, then $xy > 0$." Is the converse true? Justify your answer.

Answer: Converse: "If $xy > 0$, then $x > 0$ and $y > 0$." The converse is false.

Method: Counterexample: $x = -1, y = -1$ gives $xy = 1 > 0$, but neither x nor y is positive.

> **Investigate further:** What two cases (not simply "$x > 0$ and $y > 0$") actually follow from $xy > 0$? Day 6's OR-combinations will state this precisely.

4. True or false, with a one-line reason: "If a number's digits sum to a multiple of 3, then the number is a multiple of 3."

Answer: True.

Method: Every power of 10 leaves remainder 1 modulo 3 (since $10 \equiv 1 \pmod 3$), so a number is congruent to its digit sum modulo 3. Hence the number is a multiple of 3 exactly when its digit sum is.

> **Investigate further:** The same argument modulo 9 gives the familiar digit-sum-of-9 rule; modulo 11 it becomes an alternating sum instead. Why does the sign alternate for 11?

5. Prove directly: "If n is an odd integer, then n^2 is odd."

Answer: Proved: writing $n = 2k + 1$ gives $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$, an odd number.

Method: Let n be odd, so $n = 2k + 1$ for some integer k. Then $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$, one more than an even number, hence odd.

> **Investigate further:** Together, B2 and B5 prove both directions of "n even $\iff n^2$ even" — a full biconditional, previewing Day 2.

6. Consider: "If $x^2 = 9$ then $x = 3$." Is this true? If false, give the value of x that breaks it.

Answer: False; $x = -3$ also satisfies $x^2 = 9$.

Method: $(-3)^2 = 9$ but $-3 \neq 3$, so the implication fails for $x = -3$.

> **Investigate further:** The correct biconditional is "$x^2 = 9 \iff x = 3$ or $x = -3$" — note the OR, not a single value.

7. State the converse of Pythagoras' theorem ("if a triangle has a right angle, then $a^2 + b^2 = c^2$ for its side lengths, with c the longest"). Is this converse also a true geometric fact?

Answer: Converse: "If $a^2 + b^2 = c^2$ for a triangle's side lengths (with c the longest), then the triangle has a right angle opposite c." Yes, this converse is also true.

Method: This is the Converse of Pythagoras' Theorem: by the Law of Cosines, $c^2 = a^2 + b^2 - 2ab\cos C$, so $a^2 + b^2 = c^2$ forces $\cos C = 0$, i.e. $C = 90°$.

> **Investigate further:** This is one of the rare "both directions true" pairs — used constantly to test whether a triangle is right-angled purely from its side lengths (e.g. the 5-12-13 triangle).

8. Prove directly: "If a and b are both even integers, then $a + b$ is even."

Answer: Proved: writing $a = 2j$, $b = 2k$ gives $a + b = 2(j + k)$, an even number.

Method: Let $a = 2j$ and $b = 2k$ for integers j, k. Then $a + b = 2j + 2k = 2(j + k)$, which is even.

> **Investigate further:** Is the converse true — does "$a + b$ even" force both a, b to be even? Test $a = 1, b = 3$.

9. True or false, with a one-line reason: "If p is a prime number and $p \neq 2$, then p is odd."

Answer: True.

Method: 2 is the only even prime (every other even number ≥ 4 is divisible by 2 and hence composite), so any prime other than 2 must be odd.

> **Investigate further:** Is the converse true — is every odd number prime? Test $n = 9$.

10. State the converse of: "If $x = y$, then $x^3 = y^3$." Is the converse true?

Answer: Converse: "If $x^3 = y^3$, then $x = y$." This converse is true.

Method: Cubing is a strictly increasing (hence injective) function on the reals, so $x^3 = y^3$ forces $x = y$ — unlike squaring, cubing carries no sign ambiguity.

> **Investigate further:** Contrast with B6: squaring loses sign information (two possible roots), cubing does not (one real root). This odd-vs-even-power distinction reappears throughout the pillar.

Section C Substitution & Structure

1. Prove directly that the square of any multiple of 3 is itself a multiple of 9.

Answer: Proved: writing $n = 3k$ gives $n^2 = 9k^2$, a multiple of 9.

Method: Let $n = 3k$ for an integer k. Then $n^2 = (3k)^2 = 9k^2$, which is 9 times an integer.

> **Investigate further:** The converse ("n^2 multiple of $9 \Rightarrow n$ multiple of 3") is actually true, since 3 is prime — contrast with C8, where the analogous statement for 4 fails.

2. Prove directly: "If n is an odd integer, then $n^2 - 1$ is divisible by 8."

> **Answer:** Proved: writing $n = 2k + 1$ gives $n^2 - 1 = 4k(k + 1)$, and since $k(k + 1)$ is always even, $n^2 - 1$ is divisible by 8.

Method: Let $n = 2k + 1$. Then $n^2 - 1 = (2k + 1)^2 - 1 = 4k^2 + 4k = 4k(k + 1)$. Since k and $k + 1$ are consecutive integers, one of them is always even, so $k(k + 1) = 2m$ for some integer m. Then $n^2 - 1 = 4(2m) = 8m$, a multiple of 8.

> **Investigate further:** This is exactly why every odd square is $\equiv 1 \pmod 8$ — a workhorse identity in Number Theory Day 2's modular arithmetic toolkit.

3. Which of the following is the correct converse of: "If x is rational and y is irrational, then $x + y$ is irrational"?

> **Answer:** A) If $x + y$ is irrational, then x is rational and y is irrational.

Method: The converse of "if P then Q" is "if Q then P", formed by swapping the two clauses wholesale — never by negating them (that gives the unrelated inverse) and never by swapping only one clause. Here $P =$"x rational and y irrational", $Q =$"$x + y$ irrational", so the converse is "if Q then P", option A.

> **Investigate further:** Option C is actually the *contrapositive* of the original (negate and swap both clauses) — always logically equivalent to the original. Day 3 shows why the contrapositive, unlike the converse, is never a separate claim to prove.

4. True or false, with justification: "If a statement is true, its converse must also be false."

> **Answer:** False.

Method: B1, B7 and B10 (and A5) all exhibit true statements with true converses. One such example disproves the universal claim.

> **Investigate further:** This misconception is worth naming explicitly: reversing a conditional says nothing by itself about truth — the two directions are independent claims that sometimes happen to agree.

5. Prove directly: "If n is divisible by 6, then n^3 is divisible by 216."

> **Answer:** Proved: writing $n = 6k$ gives $n^3 = 216k^3$, a multiple of 216.

Method: Let $n = 6k$ for an integer k. Then $n^3 = (6k)^3 = 216k^3$, since $6^3 = 216$.

> **Investigate further:** More generally, if $d \mid n$ then $d^m \mid n^m$ for any positive integer m — the same substitution argument works for any power.

6. State the converse of: "For positive reals x, y: if $x \geq 2$ and $y \geq 2$, then $xy \geq x + y$." Determine whether the converse is true, giving a counterexample if it is false.

> **Answer:** Converse: "For positive reals x, y: if $xy \geq x + y$, then $x \geq 2$ and $y \geq 2$." The converse is false; $x = 1.5, y = 10$ gives $xy = 15 \geq x + y = 11.5$, but $x < 2$.

Method: Check the counterexample directly: $xy = 1.5 \times 10 = 15$ and $x + y = 11.5$, so $xy \geq x + y$ holds, yet $x = 1.5 < 2$, breaking the converse's conclusion.

> **Investigate further:** The true boundary condition is $\frac{1}{x} + \frac{1}{y} \leq 1$ (equivalent to $xy \geq x + y$ for positive x, y) — a different, correct characterisation you'll meet again in Number Theory's Egyptian-fraction questions.

7. State the converse of: "If x and y are both perfect squares, then xy is a perfect square." Is the converse true? Give a counterexample if it is false.

Answer: Converse: "If xy is a perfect square, then x and y are both perfect squares." The converse is false; $x = 2, y = 8$ gives $xy = 16 = 4^2$, but neither 2 nor 8 is a perfect square.

Method: Check directly: $2 \times 8 = 16 = 4^2$, a perfect square, while 2 and 8 are individually not perfect squares — a single counterexample disproves the converse.

> **Investigate further:** What is actually forced is that x and y's prime factorisations combine to give even exponents overall, even when neither does individually — Number Theory Day 4's prime factor algebra formalises this.

8. True or false, with justification: "If n^2 is a multiple of 4, then n is a multiple of 4."

Answer: False.

Method: $n = 6$ gives $n^2 = 36$, a multiple of 4, but 6 is not a multiple of 4.

> **Investigate further:** What is actually forced is that n is even — the correct, weaker conclusion. Contrast with C1, where the analogous statement for 3 and 9 genuinely holds, because 3's exponent bookkeeping (odd $\Rightarrow$ even after squaring $\Rightarrow$ still ≥ 1) works out differently from 2's.

Section D Challenge

1. Let n be a positive integer. Consider the statements: I. If n is divisible by 4, then n is divisible by 8. II. If n is divisible by 8, then n is divisible by 4. III. The converse of statement I is statement II. Which of the statements I, II, III are true?

Answer: F) II and III only.

Method: I: "$4 \mid n \Rightarrow 8 \mid n$" is false, since $n = 4$ is divisible by 4 but not 8. II: "$8 \mid n \Rightarrow 4 \mid n$" is true, since $8 = 4 \times 2$. III: the converse of I (swap hypothesis and conclusion of "$4 \mid n \Rightarrow 8 \mid n$") is exactly "$8 \mid n \Rightarrow 4 \mid n$", which is statement II — so III is true. Only II and III hold.

> **Investigate further:** Statement I is false but its converse (II) is true — alongside A2 and A9, a third example that a statement's truth tells you nothing about its converse's truth.

2. Prove that if n and $n + 2$ are both prime numbers greater than 3, then $n + 1$ is divisible by 6.

Answer: Proved: $n + 1$ is both even and a multiple of 3, hence a multiple of 6.

Method: Since n and $n + 2$ are primes greater than 3, both are odd, so $n + 1$ (sandwiched between two odd numbers) is even. Among the three consecutive integers $n, n + 1, n + 2$, exactly one is a multiple of 3; since n and $n + 2$ are primes greater than 3, neither is a multiple of 3, so $n + 1$ must be. Being both even and a multiple of 3, $n + 1$ is a multiple of 6.

> **Investigate further:** This is why every twin prime pair above $(3, 5)$ sits symmetrically around a multiple of 6 — e.g. $(11, 13)$ around 12, $(17, 19)$ around 18.

3. Amy claims: "If a positive integer is divisible by every one of $2, 3, 4, 5, 6$, then it is divisible by 60." Determine whether Amy's claim is true, and justify your answer using the smallest positive integer for which it can be checked.

Answer: True; the smallest positive integer divisible by all of $2, 3, 4, 5, 6$ is 60 itself.

Method: Any integer divisible by each of $2, 3, 4, 5, 6$ must be divisible by their least common multiple. Since $4 = 2^2$ and $6 = 2 \times 3$, $\mathrm{lcm}(2, 3, 4, 5, 6) = 2^2 \times 3 \times 5 = 60$. So being divisible by all five numbers is exactly equivalent to being divisible by 60, proving Amy's claim directly for every such integer, the smallest being 60 itself.

> **Investigate further:** Amy's claim is actually an "if and only if" in disguise: divisible-by-all-five and divisible-by-60 are the same condition. Day 2 gives this equivalence a name.

4. Give an example of a true statement S about divisibility of integers whose converse S' is also true. State both S and S'.

Answer: S: "If n is a multiple of 12, then n is a multiple of 4 and of 3." S' (converse): "If n is a multiple of 4 and of 3, then n is a multiple of 12." Both are true (this is exactly B1's pair). Any pair of coprime divisors d_1, d_2 works the same way.

Method: S is true because $12 = 4 \times 3$, so any multiple of 12 is automatically a multiple of both 4 and 3. S' is true because $\gcd(4, 3) = 1$, so $\mathrm{lcm}(4, 3) = 12$ divides any number divisible by both 4 and 3.

> **Investigate further:** Any answer where S's hypothesis and conclusion are logically equivalent conditions is valid — what matters is that both directions genuinely hold, not the specific numbers chosen.

5. Determine, with proof, whether the following statement is true or false: "If $n^2 - n$ is even, then n is even."

Answer: False; $n = 3$ gives $n^2 - n = 6$ (even), but n is odd.

Method: $n^2 - n = n(n - 1)$ is a product of two consecutive integers, so one of them is always even — meaning $n^2 - n$ is even for *every* integer n, regardless of n's own parity. Since the hypothesis is always true but the conclusion isn't (e.g. $n = 3$), the implication fails.

> **Investigate further:** A hypothesis that's always true makes an "if...then" statement collapse to asserting the conclusion outright — spotting this pattern (a vacuously universal hypothesis) is a recurring TMUA trap.